Monthly Notices of the ROYAL ASTRONOMICAL SOCIETY

MNRAS **499,** 3130–3138 (2020) doi:10.1093/mnras/staa2973
Advance Access publication 2020 October 2

# Classifying image sequences of astronomical transients with deep neural networks

Catalina Gómez [1]★ Mauricio Neira,[2] Marcela Hernández Hoyos,[2] Pablo Arbeláez[1]★ and Jaime E. Forero-Romero [3]★

[1]*Center for Research and Formation in Artificial Intelligence, Universidad de los Andes, Cra. 1 No. 18A-10 Bogotá, Colombia*
[2]*Systems and Computing Engineering Department, Universidad de los Andes, Cra. 1 No. 18A-10 Bogotá, Colombia*
[3]*Departamento de Física, Universidad de los Andes, Cra. 1 No. 18A-10 Bogotá, Colombia*

Accepted 2020 September 13. Received 2020 September 7; in original form 2019 December 24

## ABSTRACT

Supervised classification of temporal sequences of astronomical images into meaningful transient astrophysical phenomena has been considered a hard problem because it requires the intervention of human experts. The classifier uses the expert's knowledge to find heuristic features to process the images, for instance, by performing image subtraction or by extracting sparse information such as flux time-series, also known as light curves. We present a successful deep learning approach that learns directly from imaging data. Our method models explicitly the spatiotemporal patterns with deep convolutional neural networks and gated recurrent units. We train these deep neural networks using 1.3 million real astronomical images from the Catalina Real-Time Transient Survey to classify the sequences into five different types of astronomical transient classes. The TAO-Net (for Transient Astronomical Objects Network) architecture outperforms the results from random forest classification on light curves by 10 percentage points as measured by the F1 score for each class; the average F1 over classes goes from 45 percent with random forest classification to 55 percent with TAO-Net. This achievement with TAO-Net opens the possibility to develop new deep learning architectures for early transient detection. We make available the training data set and trained models of TAO-Net to allow for future extensions of this work.

**Key words:** methods: numerical – astronomical data bases – transients: supernovae.

## 1 INTRODUCTION

Robotic telescopes ordinarily look for transient astronomical objects of relevance such as supernovae, active galactic nuclei, asteroids, among others (Kaiser 2004; Shappee et al. 2014; Wyrzykowski et al. 2014; Smartt et al. 2015; Tonry et al. 2018; Bellm et al. 2019). The usual strategy for analyzing their output is based on obtaining temporal sequences of images on the same region of the sky and searching changes that might be of astrophysical interest. Identifying these objects are challenging due to the sparse and heterogeneous data that are captured, their evolution over time, and the sources of noise in the atmosphere or in the detectors that are inherent to observational instruments on Earth.

The astronomical community has made great advances during the last decade to perform large automated astronomical surveys aimed at finding transient objects. For instance, in 2009 the Palomar Transient Factory (PTF) managed one million transient candidates per night and determined in real-time whether each candidate was an astrophysical phenomenon of interest, or simply a detector fluctuation or a known variable source (Law et al. 2009). The recent 2019 Zwicky Transient Facility provides one order of magnitude larger data sets (Bellm et al. 2019) over the PTF. Any transient finder strives to achieve a high recall to avoid missing interesting events, and a high precision to reduce the number of false alarms (Bloom et al. 2012). Although the image subtraction algorithm is an accurate and fast strategy for separating transient sources from non-transients, the fine-grained classification into transient categories demands more complex models that learn richer representations. Both the robust classification and follow-up decisions are key to harness the potential of forthcoming sky surveys. These requirements have been typically met by having expert input to decide on the image features that should be the most relevant for an algorithm to make the classification.

Fully general algorithms that take as input a sequence of observed images to detect and learn both the spatial and temporal features relevant for classification into astronomical classes are still in their infancy. Previous efforts were able to find features in images that could be classified as point-like, streak-like or artefacts (Jia et al. 2019), distinguishing real supernovae candidates from bogus events (Cabrera-Vives et al. 2016) or training algorithms with simulated images to classify a few hundred observed image sequences (Carrasco-Davis et al. 2019).

In this paper, we propose to model the spatiotemporal nature of the problem through modern recognition techniques. To do so, we retrieve more than 1 million real images for multiple transient objects identified in the Catalina Real-Time Transient Survey (CRTS) catalogue. We choose this survey because, in contrast to all the other surveys above, its transient discoveries are public, and the original images can be retrieved from the catalogue in addition to metadata for each object. The image data set that we retrieve reflects the

★ E-mail: c.gomez10@uniandes.edu.co (CG); pa.arbelaez@uniandes.edu.co (PA); je.forero@uniandes.edu.co (JEF-R)

© 2020 The Author(s)
Published by Oxford University Press on behalf of the Royal Astronomical Society

Downloaded from https://academic.oup.com/mnras/article/499/3/3130/5917433 by guest on 15 October 2024

imbalanced problem of identifying transient objects and the challenges of capturing data of objects that are beyond our atmosphere. Our approach leverages both the spatial and temporal information to improve the classification of nearly 7000 image sequence into different categories of astronomical transients, compared to the machine-learning algorithms based on light curves and hand-crafted features.

We expect our work to contribute to the development of robust algorithms for future transient surveys such as the Large Synoptic Survey Telescope (Ivezic et al. 2009), expected to revolutionize time-domain astronomy (by characterizing several millions of transients every night, gathering 15 terabytes of data every night) after it comes online in 2020.

This paper is structured as follows. In Section 2, we describe the image data set and its acquisition process, and the classification tasks that we define. In Section 3, we present the light curves approach and the deep learning architecture that learns directly from the image sequences. Then, in Section 4, we present the performance of our model for the different tasks that we propose, and finally, in Section 5, we expose concluding remarks of our work.

## 2 TRANSIENT ASTRONOMICAL OBJECT IMAGE DATA SET

We retrieve the images of TAO data set from the public catalogues of the CRTS (Drake et al. 2009; Mahabal et al. 2011), a survey that looked for highly variable objects. The area covered by the CRTS is 33 000 deg$^2$. The project has been taking data since 2007 with three telescopes: Mt. Lemmon Survey, Catalina Sky Survey (CSS), and Siding Spring Survey. CRTS has reported more than 15 000 transient events. The data we use comes from the CSS telescope located in the Santa Catalina Mountains in Arizona. The images come from a 111-megapixel camera measuring in the *V*-Band. This telescope and its detector have a scale of 2.5 arcsec per pixel.

### 2.1 Data set acquisition

We use five transient classes from the public by CRTS that include five classes: blazars (BZ), active galactic nucleus (AGN) event, cataclysmic variables (CV), supernovae (SN), and other objects that include events of unknown nature (OTHER) (Drake et al. 2009). Each object has information about its right ascension, declination, magnitude, discovery date, classification, and light curve points. Fig. 1 shows a series of sample images from this data set for all the six different transient classes.

The survey released the light curves of the transient objects, and recently, the third data release includes cutout images of the objects identified by the CSS using the Schmidt telescope from the period between 2003 to 2012. We build the series of images over a window of three years, where the second year always includes the date of maximum brightness. The released cutouts are cropped from an original image (of size 4110 × 4096 pixels). They include information about the cutout matrix location, the date in which the image was captured, the field identifier and the observation number. However, having an image sequence centred on a given location on the sky is not trivial. We had to implement web scraping techniques to access and reconstruct the images for each transient sequence. The web query also gives us the corresponding light curve for the transient of interest.

The CRTS public catalogues only include transient objects, but most of the objects in a survey are non-transient. To mimic the imbalanced nature of the detection problem, we retrieve more

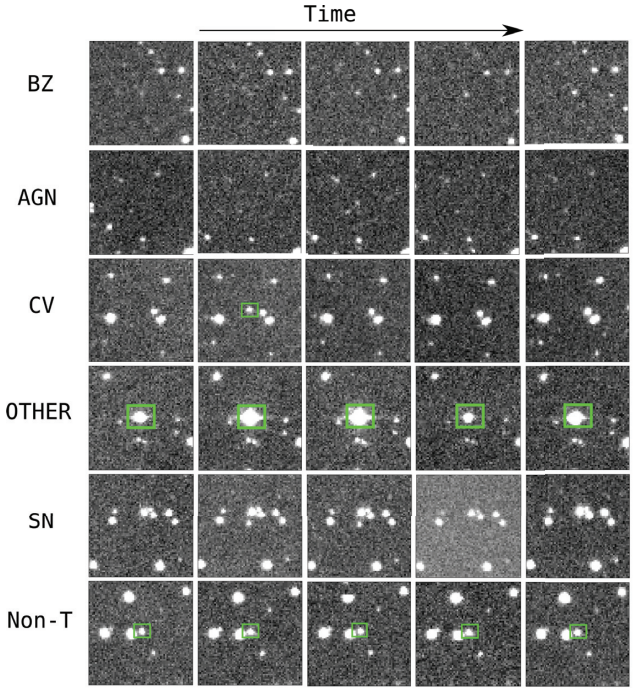



**Figure 1.** Sample images in the data set. Each row corresponds to a sample of a different class. The temporal spacing between consecutive images varies for each example. The transient is always in the centre of the image. Every image has 64 × 64 pixels corresponding to an angular size of 2.7 arcmin. Images are normalized for visualization.

examples of non-transient objects by selecting a different light source from a reference cutout. To search for image samples of non-transient objects, we use the original cutouts of sources that were present during the three years but did not have any transient associated to them. In general, these non-transient sources do not have an associated light curve. The non-transient light curves are not available. The complete search of images to create the training data set took about 11000 CPU-hours.

### 2.2 Data base Description

We create a data base that comprises image sequences for transient and non-transient objects. The raw FITS images are two dimensional (120 × 120 pixels). Each Region of Interest (RoI) has a size of 64 × 64 pixels and preserves the original values of the FITS images. Example sequences of transients and non-transient objects are shown in Fig. 1. Due to the original imbalance of transient classes, our data set comprises more instances and images for some classes. The annotation for each image is provided by the object type registered in the catalogue. In Table 1, we show the statistics concerning the class distribution, the number of cutouts that we download, and the final number of images for each category. In the final count of objects that we download (*Downloads*), we ensure that each one has images at the date of maximum brightness, such that the three-year observation period captures any brightness variation.

### 2.3 Data base Realism

Our data set captures the three most important challenges in transient classification as presented in a survey. First, each transient type has a different brightness behaviour that changes non-periodically over time. Besides, the temporal scale at which these changes occur is

Downloaded from https://academic.oup.com/mnras/article/499/3/3130/5917433 by guest on 15 October 2024

**Table 1.** General statistics of TAO data set for transient and non-transient images retrieved over the three-year observation period. The count *In catalogue* corresponds to the transients count of CRTS, *Downloads* to the number of objects that have observations at the date of maximum brightness, *Cutouts* is the total count of images that we download from the survey, and *RoIs* is the image count with centred objects.

| Count | BZ | AGN | CV | OTHER | SN | Transients | Non-transients | Total |
|---|---|---|---|---|---|---|---|---|
| In catalogue | 270 | 651 | 987 | 1054 | 1723 | 4712 | – | 4712 |
| Downloads | 239 | 606 | 776 | 821 | 1372 | 3838 | 14 817 | 18 655 |
| Cutouts | 23 480 | 67 034 | 74 703 | 75 257 | 148 082 | 390 659 | 1106 921 | 1497 580 |
| RoIs | 22 281 | 64 576 | 65 852 | 73 092 | 137 475 | 363 276 | 1028 358 | 1391 634 |

**Table 2.** Number of objects per classes in the fixed split for training and validation.

| Set | BZ | AGN | CV | OTHER | SN | Non-transient | Total |
|---|---|---|---|---|---|---|---|
| Train | 157 | 413 | 505 | 564 | 916 | 9168 | 11 723 |
| Validation | 68 | 177 | 217 | 242 | 394 | 5649 | 6747 |
| Test | 14 | 28 | 54 | 38 | 62 | 682 | 878 |

not homogeneous, and presents intraclass and interclass variation. Secondly, we build the non-transient data base to have similar brightness across time as the transient objects. This suppresses a first-order difference that would make the classification of non-transients easier. Thirdly, the sampling of images during the three-year period is non-uniform for all the transients; intervals range from days to months, and the number of observations at different dates varies within classes.

### 2.4 Classification tasks

We define four classification tasks characterized by a large class imbalance. The first is transient classification, that is, classifying transients versus non-transients. The second one classifies SN versus non-transients. The third is a multiclass detection that involves separating transients from non-transients, and finally a fine-grained classification of the five transient classes.

We evaluate all tasks with the metrics of a detection problem due to the large class imbalance. For each class, we report the maximum F1 score (F1) from the Precision–Recall curve that we construct by setting different thresholds on the output probabilities of each class. We report the metrics for each class as the average F1 score of all the objects within the class. The global performance is the average of the metrics for individual classes plus or minus the standard deviation.

### 2.5 Data set splits

For our experimental framework, we define a fixed partition to train and validate the models. We start by discarding sequences of transient objects where we could not recover the complete sequence of images (missing observations at the date of maximum brightness) or at least three observations. We also remove six transient objects that had uncertain classes (labelled as both BZ and AGN). Then, we randomly select the 70 per cent of instances for each class as the training set, and the remaining 30 per cent as the validation set, keeping the class distribution similar. For the non-transient class, we select the 60 per cent of objects for training models. In addition, we define a separate test set for the final evaluation with less objects for each category.

Table 2 summarizes the partitions for individual classes, and reflects the imbalance of transients compared to non-transients (1:5 ratio). Likewise, the ratio of samples at each transient class to non-transient class ranges between 1:14 for supernovae, and 1:81 for blazars.

## 3 METHODS

### 3.1 Random forest on light curves

We implement a traditional approach to transient classification using the *V*-band light curves compiled from the CRTS catalogue that corresponds to the transient objects that we retrieved. This is the baseline against which we compare our deep learning implementation. Unfortunately, the structure of the data base does not allow us to construct the corresponding light curve catalogue for non-transients. Therefore, the complete comparison between image sequences and light curves can only be done for the classification into five transient classes.

We compute the discriminatory features reported by Richards et al. (2011) and D'Isanto et al. (2016) over the light curves. These features fall into three categories: moment based, percentile based, and magnitude/flux based. For further details into the definition of features, please see the appendix of Richards et al. (2011).

To balance transient classes, we generate additional examples for less frequent classes. We define a sampling strategy in which we sample a Gaussian probability distribution with the magnitude as the mean and the corresponding error as the standard deviation. This technique generates new light curves that are slightly different from the same number of observations as the original. We balance the set such that all classes have the same number of instances of the most represented class (9168 instances for each class, to match the number of non-transients). Additionally, we filter out the objects with less than five observation dates to avoid errors in the feature estimation.

After calculating the features and balancing the classes, we use them with the annotations to train a random forest classifier, as in Richards et al. (2011), with 200 trees. We also explore other classifiers such as support vector machines and neural networks and find that the best results are obtained with a random forest classifier, as reported in the experimental comparison of Neira et al. (2020).

### 3.2 TAO-Net: our neural network architecture

The unified neural network architecture, we propose (TAO-Net, for transient astronomical object) has two main components as sketched in Fig. 2. It consists of two modules. First, we use CNNs to extract a feature representation from the image sequences, and then, encode the sequence of features with recurrent neural networks (RNN). We train models from scratch due to significant differences with natural images.



Downloaded from https://academic.oup.com/mnras/article/499/3/3130/5917433 by guest on 15 October 2024

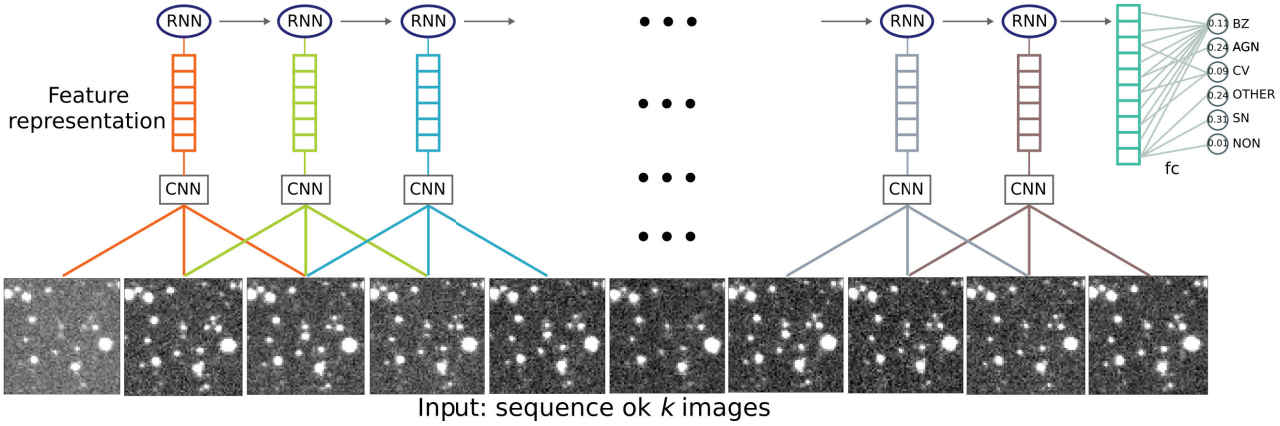



**Figure 2.** Overview of TAO-Net. The model takes the raw sequence of images as the input for consecutive CNNs to create high-level features for a recurrent module that analyzes the whole image sequence. We map the output of the RNN network to a probability distribution over the classes.

Deep neural networks can integrate low-/mid-/high-level features with classifiers for large-scale visual recognition tasks. We experiment with the state-of-the-art densely connected convolutional networks (DenseNet) because of their improved flow of information and gradients throughout the network, the feature reuse at different layers and deep supervision (Huang et al. 2017). These networks comprise $L$ layers, and define the growth rate $k$ parameter to control the contribution of each layer to the global state of the network. We train DenseNet with different depths and growth rates.

Since we are processing sequential data, we explore two variants for fusing information over the temporal domain as in Karpathy et al. (2014). The early fusion model combines the temporal information before the first convolutional layer, namely, at the pixel level. Conversely, the late fusion strategy requires as many networks (with shared parameters) as inputs, and then merges the streams before the classification layer. With a SoftMax function, we map the output scores to a class probability distribution.

To model complete image sequences and learn long-term dependencies, we include an RNN network over the DenseNet network. Two particular models, long-short term memory cells and gated recurrent units (Cho et al. 2014) have been successful for processing sequence data. We experiment with GRUs because of their efficiency, both at training and testing. In these models, each layer $L$ has a hidden state of dimension $H$ at each time-step of the sequence. We evaluate different numbers of layers, dimensions and a bidirectional structure.

To generate the input sequence for the GRU, we split the image sequence every three consecutive dates and use the triplet as the input for DenseNet without the classification layer to create a sequence of features. The GRU encodes and decodes the sequence, and then we include a fully connected layer to generate the class scores. The complete model, TAO-Net, includes both the DenseNet and GRUs.

We include temporal information from the non-uniform sampling by concatenating the sequence of relative dates (difference between consecutive dates in years) to the sequence of features generated with the DenseNet model, and use these new features as input for the GRU units.

### 3.3 Training details

We first train our models in the multiclass problem, and once the model has learned to recognize transient classes, we use a transfer learning strategy called finetuning (Yosinski et al. 2014) to adjust the model's weights for binary tasks, such as supernovae and transient detection. The weights of the pretrained model are said to be finetuned by continuing backpropagation in a new set of data with a low learning rate. All the layers from the network can be finetuned, or only a fraction by freezing the weights of the layers that want to be preserved.

We begin training DenseNets and GRUs independently. To model the temporal information in DenseNets, we sample images from the complete sequences at three different dates in sequential order, such that they reflect differences in brightness for transient classes. We include the observation date in the three year period when the transient object had the maximum brightness, and one observation before and after that date. For the non-transient class we take the first, middle and last dates of the sequence of ordered images. After training DenseNet with three images as input to discriminate among transient and non-transient categories, we freeze the weights of DenseNet's layers, and fix the feature representation of the complete sequences to train the GRU network apart. We adjust the training protocol to process variable-length sequences, since each object has a different number of sequential observations and we aim at including all the temporal information.

To alleviate class imbalance, we exploit the multiple observations on the same date to generate a slightly different sequence to represent each object. To do so, we define a permutation with the multiple observations at each date and over-ordered dates, changing at least one observation at any date. We adjust the number of new sequences for each object to ensure that we have balanced classes on the training set.

We further improve our results when we train TAO-Net's components together, called joint training, adjusting the weights of the DenseNet to generate the feature sequences and the weights of the recurrent units. To ensure that the features are accurate for the final prediction, we include an intermediate error function after the final DenseNet layer, and add this loss with the one computed for the model's final predictions. For the joint training, we set the sequence length of all instances to a fixed value $S$ by taking the central observations if the number of observations was greater than $S$, or by replicating the first and last observations such that the new length is $S$. We define $S = 19$ for all the experiments because of memory limitations.

## 4 RESULTS

Table 3 summarizes the results for all the classification tasks in the validation and test sets. We use all the data in the training and

Downloaded from https://academic.oup.com/mnras/article/499/3/3130/5917433 by guest on 15 October 2024

**Table 3.** Performance comparison in terms of F1-measure in the four classification tasks. The mean value and standard deviation are computed over the classes. For the classification task into five transient classes, TAO-Net presents an average F1 score almost nine points higher than the results based on light curves and a random forest classifier.

| Classification task | Data set | ML model | F1 validation | F1 test |
|---|---|---|---|---|
| Transient/non-transient | Images | TAO-Net | 89.31 ± 6.98 | 92.50 ± 4.12 |
| SN/non-transient | Images | TAO-Net | 84.90 ± 12.79 | 86.51 ± 10.99 |
| 5 Transients (Blazar, AGN, cataclysmic variables, supernovae, and other) and non-transient | Images | TAO-Net | 53.78 ± 22.31 | 51.51 ± 20.56 |
| 5 transients | Light curves | Random Forest | 45.49 ± 13.75 | 46.65 ± 10.47 |
| | Images | TAO-Net | 54.58 ± 13.32 | 55.31 ± 10.08 |

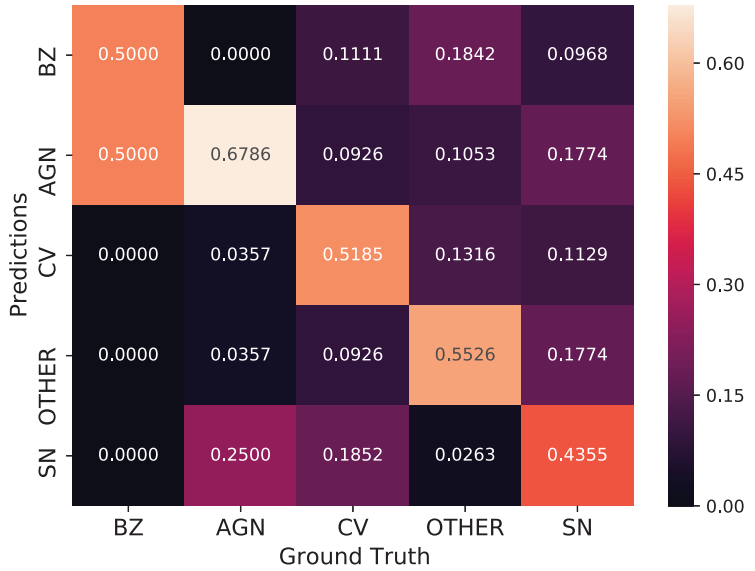



**Figure 3.** Confusion matrix of the predictions generated with TAO-Net double supervision model in the test set for the five transient classes classification.

**Table 4.** Performance comparison in terms of F1 score for the binary detection problem. Each row corresponds to a different experiment, and the last column reports the average F1 score of both classes.

| Data | Model | Transient | non-transient | F1($\mu \pm \sigma$) |
|---|---|---|---|---|
| Images | DenseNet, $k = 32$, $L = 70$ | 74.46 | 95.06 | 84.76 ± 10.30 |
| Images | TAO-Net: (DenseNet, $k = 32$, $L = 70$) + (GRU, $L = 2$, $H = 128$) | 78.56 | 95.56 | 87.06 ± 8.5 |
| Images | TAO-Net w/ two loss[a], $S = 19$ | 82.38 | 96.22 | 89.30 ± 6.92 |

[a]All weights of TAO-Net were updated during the joint training.

validation subsets to retrain the best models found on the latter set. Overall, we obtain similar F1 scores in the test and validation data sets. In the five transients task, TAO-Net achieves an average F1 score of 55.31 ± 10.08, almost nine points higher than the result from the random forest classification on light curves (46.65 ± 10.47). Furthermore, TAO-Net achieves a higher F1 score for every class. We summarize the performance of each category in the test set under the different tasks in Appendix A in Tables A1–A3. For instance, while in TAO-Net the supernovae achieve the highest score (63.75) and Blazars achieve the lowest (36.84), the random forest scores are 57.15 and 32.00, respectively.

The confusion matrix in Fig. 3 shows that supernovae are most easily misclassified as active galactic nuclei and other objects, while blazars are confused with AGNs as well. Conversely, AGNs and CVs classes are most commonly misclassified as supernova. In the next subsection, we present the detailed F1 scores for each class and different TAO-Net configurations in the validation set.

### 4.1 Binary Problem

For the binary experiments, we merge the instances from all the transient classes and apply the balance strategy using multiple observations. We compare the DenseNet model using only three images as input and GRUs with the complete image sequences. Table 4 summarizes the F1 score for different models in the binary task. We observe that with DenseNet one achieves a satisfactory result to detect transient objects using only three images in sequential order. To evaluate the performance TAO-Net, we first freeze the learned weights of DenseNet's layers to train the GRU network apart, and then, we jointly learn all the weights of TAO-Net with the intermediate supervision. The performance of the former is shown in the second row of Fig. 4, and the latter in the last row. The best model for the binary task is TAO-Net in the joint training and double supervision. The confusion matrix of this model is shown in Fig. 4. There is an absolute improvement of eight points in F1



Downloaded from https://academic.oup.com/mnras/article/499/3/3130/5917433 by guest on 15 October 2024

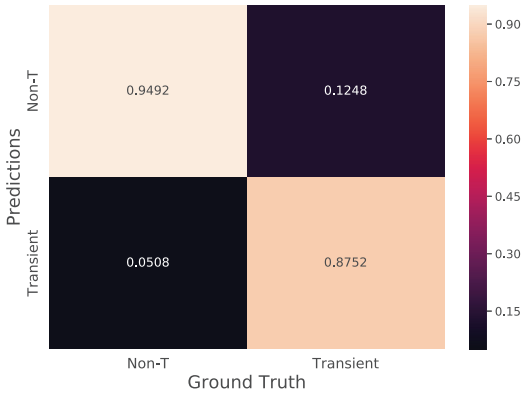



**Figure 4.** Confusion matrix of the predictions generated with TAO-Net double supervision model in the transients versus non-transients configuration.

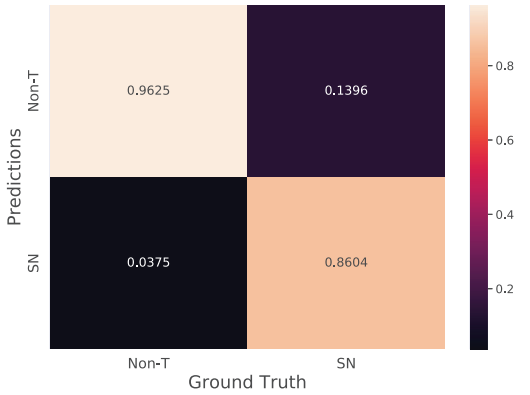



**Figure 5.** Confusion Matrix of the predictions generated with TAO-Net double supervision model in the SN versus non-transients configuration.

for the transient class and, to a lesser extent, 1.16 points, for the non-transient class.

### 4.2 Supernovae versus non-transients

In Table 5, we show the results of the binary supernovae classification problem. We initialize TAO-Net from the weights learned for the multiclass problem. We note that the performance of the non-Transient class does not change when using different model configurations. Similarly, the F1 score of the supernovae class presents a slight improvement when we update the weights of the convolutional module with double supervision, but fix the weights of the GRU units, instead of finetuning the complete model. The confusion matrix of the best model for SN classification is shown in Fig. 5.

### 4.3 Five transient classes and non-transients

Table 6 shows the results of transient classification including the non-transient category. Overall, we note that the average performance increases when we add the long-term temporal information, first when we include it as three sequential images, and then as complete sequences. For the TAO-Net experiments within the top block of the table, we fix the weights of the DenseNet model with the best configuration (first row) to calculate the sequence of features.

The average performance, shown in the last column, increases when we add the long-term temporal information, first when we include it as three images in sequential order, and then as complete sequences. When we include the temporal information as relative dates in the sequence of features (denoted as TAO-Net, dates, FT in Table 6), we finetune TAO-Net model but do not observe an improvement in the individual and average performance.

For the joint training of TAO-Net (bottom block of Table 6), we initialize the weights from the best TAO-Net model (second row). We first fix the weights of the RNN module, and learn more appropriate DenseNet features for the final classification with the intermediate supervision. In this configuration (TAO-Net w/ 2 loss), the average performance increases in almost two points with a slight improvement for individual classes, except for the OTHER class in which the F1 score increased almost 12 points. We also re-train all the weights of the network (TAO-Net w/ 2 loss[a]), in which we gain almost four points in the average performance, and achieve the best F1 score for the less represented classes (BZ, AGN, and CV). The confusion matrix for all the classes is shown in Fig. 6. First, we observe that it is not common for the model to predict any transient class for non-transient objects, which is reflected in the high performance for this class. Secondly, transients are confused with other transient types, such as AGNs (0.39) and cataclysmic variables (0.24) are more commonly confused with supernovae. Besides, all transient categories present at least 10 per cent of their instances confused with the non-transient category, especially the other objects class (0.25).

It is worth noting that, for blazars, AGNs and cataclysmic variables, the performance is proportional to the representation of the class in the data set. Regarding the 'Other' objects class, there is a large variability within the class because of the heterogeneous nature of events that were assigned to this category, which is reflected in a low F1 score.

### 4.4 Five transient classes

In Table 7, we compare the performance of our approach based on image sequences to the random forest model Neira et al. (2020) with features from the magnitude light curves provided by the catalogue. This comparison is limited to the classification task within transient categories, since we do not have CRTS light curves for the non-transient sources that we retrieved. When we include temporal information as three sequential images, we observe an absolute improvement in the average F1 score of almost seven points, and an increasing performance for individual classes, with respect to the light curves experiment. In the experiments with TAO-Net, in the last two rows of Table 7, we further improve the performance for all

**Table 5.** Performance comparison in terms of F1 score for the binary supernova problem. Each row corresponds to a different experiment, and the last column reports the average F1 score of both classes.

| Data | Model | Supernova | Non-transient | F1($\mu \pm \sigma$) |
|---|---|---|---|---|
| Images | DenseNet, $k = 32$, $L = 70$ | 71.96 | 97.88 | $84.92 \pm 12.96$ |
| Images | TAO-Net w/ two loss, $S = 19$ | 72.10 | 97.69 | $84.90 \pm 12.79$ |

Downloaded from https://academic.oup.com/mnras/article/499/3/3130/5917433 by guest on 15 October 2024

**Table 6.** F1 score for each class in the multiclass detection. Each row corresponds to a different experiment, and the last column reports the average F1 score of the 6 classes.

| Data | Model | BZ | AGN | CV | OTHER | SN | Non-transient | F1($\mu \pm \sigma$) |
|---|---|---|---|---|---|---|---|---|
| Images | DenseNet, $k = 32$, $L = 70$ | 21.82 | 37.45 | 54.76 | 40.22 | 46.59 | 95.29 | $49.36 \pm 22.84$ |
| Images | TAO-Net | 18.70 | 41.04 | 56.06 | 36.18 | 53.04 | 95.33 | $50.06 \pm 23.64$ |
| Images | TAO-Net, dates, FT | 19.15 | 36.76 | 55.86 | 37.66 | 50.69 | 95.24 | $49.23 \pm 23.66$ |
| Images | TAO-Net w/ 2 loss | 20.17 | 42.22 | 58.44 | 48.14 | 50.53 | 95.64 | $52.52 \pm 22.62$ |
| Images | TAO-Net w/ 2 loss[a] | 22.09 | 45.87 | 63.71 | 46.05 | 49.43 | 95.52 | $53.78 \pm 22.31$ |

[a]All weights of TAO-Net were updated during the joint training.

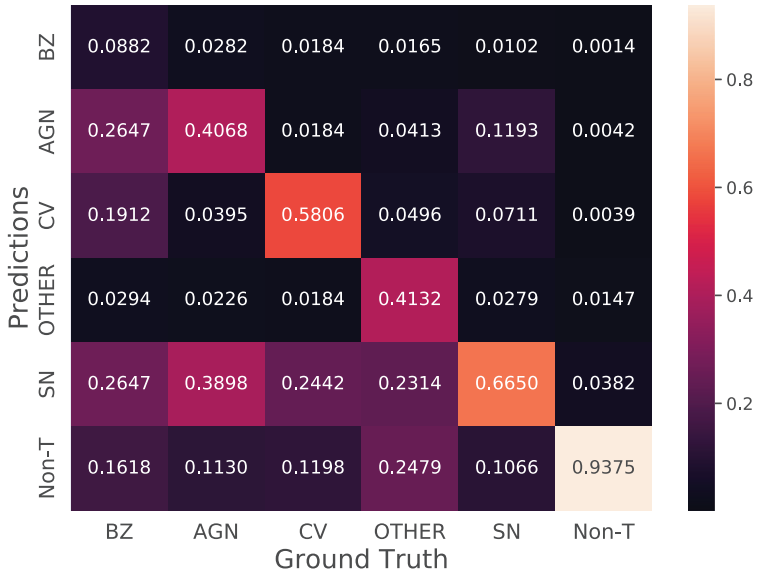



**Figure 6.** Confusion matrix of the predictions generated with TAO-Net double supervision model in the multiclass configuration.

**Table 7.** F1 score for each class in the transient classification. Each row corresponds to a different experiment, and the last column reports the average F1 score of the five transient categories. TAO-Net outperforms the random forest classifier by almost 10 percentage points in every class.

| Data | Model | BZ | AGN | CV | OTHER | SN | F1($\mu \pm \sigma$) |
|---|---|---|---|---|---|---|---|
| Light curves | Random forest | 19.74 | 42.67 | 53.60 | 56.06 | 55.36 | $45.49 \pm 13.75$ |
| Images | DenseNet, $k = 32$, $L = 70$ | 25.17 | 49.77 | 59.48 | 64.04 | 63.39 | $52.37 \pm 14.53$ |
| Images | TAO-Net | 28.57 | 47.48 | 58.43 | 65.58 | 67.08 | $53.43 \pm 14.23$ |
| Images | TAO-Net w/ 2 loss | 29.29 | 53.02 | 62.93 | 62.39 | 65.28 | $54.58 \pm 13.32$ |

transient classes. We achieve the best average performance when we jointly train TAO-Net with double supervision (w/ 2 loss).

The confusion matrix with the predictions for all transient categories from the best TAO-Net model is shown in Fig. 7. Blazars are most commonly confused with three transient categories: AGNs, other objects, and cataclysmic variables. The AGN class presents the greatest confusion with supernovae (and vice versa), and CVs are confused with supernovae too, but also with other objects to a lesser extent.

If we compare the performance of transient classes to the results in the classification setup with the non-transient category described in Section 4.3, we notice that the metrics are better for all transient classes, except cataclysmic variables, for which the performance decreases only one point. We attribute the lower performance in the six-class setup to the common confusions of the non-transient class with transient categories, such as blazars and other objects, as shown in the confusion matrix from Fig. 6.

### 4.5 Ablation Experiments

We perform different ablation experiments that corroborate our design choices for TAO-Net. First, we analyse the effect of changing $k$ and $L$ of DenseNet models with an early fusion strategy. Once we define the best combination of $k$ and $L$, we explore the late fusion model. Table 8 compares DenseNet's performance under different configurations. The large model with $k = 64$ and $L = 70$ layers does not improve the performance over TAO-Net. A possible explanation is that the model has four times the number of parameters of TAO-Net (3.1M) and, since we train the models from scratch, the number of training instances does not suffice. The DenseNet model with 90-layers follows a similar pattern, but with a higher performance drop.

We also reduce the model complexity, first by changing the number of layers to $L = 50$, which presents considerable underfitting and underperforms with respect to TAO-Net. Second, the model with a smaller growth rate $k = 16$ underfits the data and reduces the average performance significantly.



Downloaded from https://academic.oup.com/mnras/article/499/3/3130/5917433 by guest on 15 October 2024

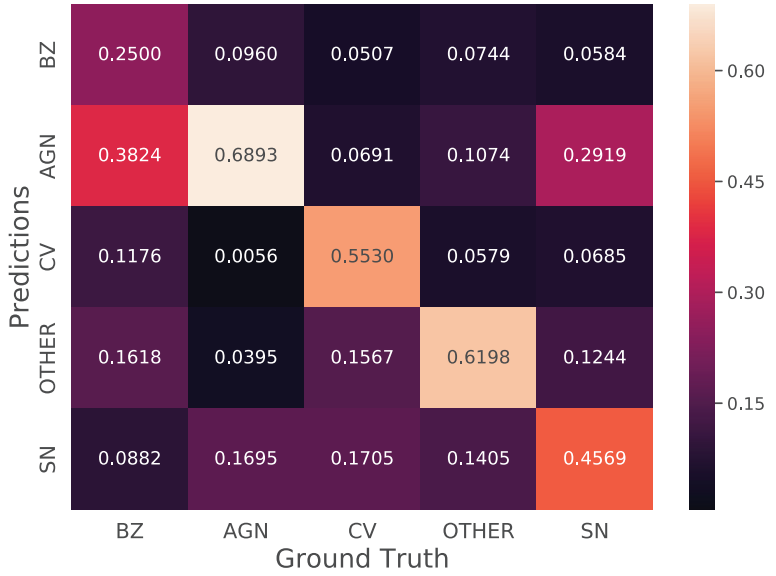



**Figure 7.** Confusion matrix of the predictions generated with TAO-Net double supervision model for the five transient classes classification.

**Table 8.** Results in the ablation experiments of DenseNet for the six-class problem. We report the mean and standard deviation of the F1 score over the six classes.

| Model | Configuration | F1 ($\mu \pm \sigma$) |
|---|---|---|
| DenseNet | $k = 32, L = 70$ | $49.36 \pm 22.84$ |
| DenseNet | $k = 64, L = 70$ | $42.35 \pm 25.27$ |
| DenseNet | $k = 32, L = 90$ | $39.41 \pm 25.60$ |
| DenseNet | $k = 32, L = 50$ | $29.41 \pm 28.19$ |
| DenseNet | $k = 16, L = 70$ | $26.98 \pm 29.01$ |
| DenseNet – Late Fusion | $k = 32, L = 70$ | $46.17 \pm 25.03$ |

**Table 9.** Results in the ablation experiments of RNN for the six-class problem. We report the mean and standard deviation of the F1 score over the six classes. In the models, *last* means that we take the last element from the output sequence, and *init* that we use pretrained weights on the binary problem.

| Model | Configuration | F1 ($\mu \pm \sigma$) |
|---|---|---|
| GRU-last | $H = 128, L = 2$ | $48.21 \pm 24.11$ |
| GRU-last | $H = 256, L = 2$ | $47.93 \pm 24.21$ |
| BiGRU-last | $H = 128, L = 2$ | $49.01 \pm 23.15$ |
| GRU-init-LR | $H = 128, L = 2$ | $49.27 \pm 23.37$ |
| GRU-add | $H = 128, L = 2$ | $50.06 \pm 23.64$ |

We find $k = 32$ and $L = 70$ to be the best DenseNet configuration, and explore the late fusion strategy. The average performance decreases about three points when compared to DenseNet with early fusion. An advantage of using an early fusion approach is that there is a direct connectivity to the pixel data, and the network can detect local variations (Karpathy et al. 2014).

Finally, we use TAO-Net to generate a fixed high-level feature representation, and evaluate the contribution of modelling sequential information with RNNs.

We then compare different hyperparameters, weight initialization and output processing of the GRU models. Table 9 shows the ablation tests and results for different RNN configurations. Overall, we observe that the average performance does not change significantly when we modify the model configuration. We evaluate two variants for processing the output of the GRU. The first one takes the last element of the sequence (*last*) and uses it as the input of the classification layer, while the other one aggregates the elements of the output sequence in the temporal dimension (*add*). We find a small increase when adding the information of the complete sequence.

**Table 10.** Experiments varying the input size of DenseNet for the Five Transient Classes. We report the mean and standard deviation of the F1 score over the five classes in the validation set.

| Model | Configuration | F1 ($\mu \pm \sigma$) |
|---|---|---|
| RF | trees = 200 | $45.49 \pm 13.75$ |
| DenseNet – 64pix | $k = 32, L = 70$ | $52.37 \pm 14.53$ |
| DenseNet – 32pix | $k = 32, L = 70$ | $51.92 \pm 15.28$ |
| DenseNet – 16pix | $k = 32, L = 70$ | $51.44 \pm 15.75$ |

Besides, we explore two larger models, increasing the dimension of the hidden state to $H = 256$ and a bidirectional GRU, the latter yielding better results.

To explore the benefit of learning from pretrained weights, we initialize the model with weights from the equivalent model that was trained for the binary task, such that it has already learned to distinguish transients from non-transients. We use the best binary model from Table 4 to define the initialization and apply a learning rate reduction every 30 epochs, which gives a slight increase in performance.

In addition, for the best DenseNet model in the five transients setup, we reduce the image size of the sequences by cropping the original images centred at the object at each observation date. The performance of the model when changing the input size (32pix and 16pix in Table 10) demonstrates that including more context improves the F1 score, and is further better than the light curves with the hand-crafted features and the Random Forest classifier (first row).

## 5 CONCLUSIONS

In this paper, we presented a fully observation-driven classification algorithm that learns the spatial and temporal patterns to assign a

Downloaded from https://academic.oup.com/mnras/article/499/3/3130/5917433 by guest on 15 October 2024

sequence of images to a transient category. We validated our approach in the data set that we retrieved from the image sequences acquired by the telescopes of the CRTS. This data base is of unprecedented realism as it reflects the inherent challenges of identifying transient astronomical objects.

The success of our imaged-based approach to classification showcases the potential of deep learning to augment expert astronomical knowledge to extract relevant spatial and temporal features as a complement to what is offered by light curves and image subtraction. The achievement of the TAO-Net architecture opens different ways for future work. For instance, while TAO-Net has been tested with single-filter images, it is possible to extend it and include information from other filters. Another promising extension, given the success in classifying SN, is to develop TAO-Net for early SN detection or more generally as a broker to provide real-time triggers for transient follow-up.

Finally, to provide the possibility to fully reproduce our results and extend our work based on observational data, we make publicly available our training data set, source code and trained models of TAO-Net. This experimental framework will allow detailed comparisons against future deep learning architectures or machine-learning methods that use light curves as an input (Charnock & Moss 2017; Ishida et al. 2019; Möller & de Boissière 2020; Pasquet et al. 2019).

## ACKNOWLEDGEMENTS

The authors thank the Office of the Vice Rector for Research at the Universidad de los Andes for supporting this project by the grant SPATIO TEMPORAL TRANSIENT OBJECT / P17.246622.004/01.

## 6 DATA AVAILABILITY

The data underlying this paper are available through https://crts.iucaa.in/CRTS/ (Drake et al. 2009). The complete set of reconstructed and annotated images will be available in a repository. The training data set and the trained model are available upon request.

## REFERENCES

Bellm E. C. et al., 2019, PASP, 131, 018002
Bloom J. S. et al., 2012, PASP, 124, 1175
Cabrera-Vives G., Reyes I., Förster F., Estévez P. A., Maureira J.-C., 2016, International Joint Conference on Neural Networks (IJCNN), p. 251
Carrasco-Davis R. et al., 2019, PASP, 131, 108006
Charnock T., Moss A., 2017, ApJ, 837, L28
Cho K., van Merrienboer B., Gülçehre Ç., Bougares F., Schwenk H., Bengio Y., 2014, EMNLP, 1724
D'Isanto A., Cavuoti S., Brescia M., Donalek C., Longo G., Riccio G., Djorgovski S. G., 2016, MNRAS, 457, 3119
Drake A. et al., 2009, ApJ, 696, 870
Huang G., Liu Z., Van Der Maaten L., Weinberger K. Q., 2017, in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, p. 4700
Ishida E. E. O. et al., 2019, MNRAS, 483, 2
Ivezic Z. et al., 2009, AAS, 213
Jia P., Zhao Y., Xue G., Cai D., 2019, AJ, 157, 250
Kaiser N., 2004, in Oschmann Jacobus M. J., ed., Proc. SPIE Conf. Ser. Vol. 5489, Ground-based Telescopes, SPIE, Bellingham, p. 11
Karpathy A., Toderici G., Shetty S., Leung T., Sukthankar R., Fei-Fei L., 2014, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, IEEE, p. 1725
Law N. M. et al., 2009, PASP, 121, 1395
Mahabal A. A. et al., 2011, Bull. Astron. Soc. India, 39, 387
Möller A., de Boissière T., 2020, MNRAS, 491, 4277
Neira M., Gómez C., Suárez-Pérez J. F., Gómez D. A., Reyes J. P., Hoyos M. H., Arbeláez P., Forero-Romero J. E., 2020, ApJS, 250, 11
Pasquet J., Pasquet J., Chaumont M., Fouchez D., 2019, A&A, 627, A21
Richards J. W. et al., 2011, ApJ, 733, 10
Shappee B. J. et al., 2014, ApJ, 788, 48
Smartt S. et al., 2015, A&A, 579, A40
Tonry J. L. et al., 2018, PASP, 130, 064505
Wyrzykowski Ł. et al., 2014, Acta Astronomica, 64, 197
Yosinski J., Clune J., Bengio Y., Lipson H., 2014, Advances in neural information processing systems, 3320

## APPENDIX A: PERFORMANCE ON THE TEST SET

**Table A1.** F1 score for each class in the test set for transient classification. The last column reports the average F1 score of the five transient categories.

| Set | Data | Model | BZ | AGN | CV | OTHER | SN | F1($\mu \pm \sigma$) |
|---|---|---|---|---|---|---|---|---|
| Test | Light curves | RF | 32.00 | 36.00 | 54.54 | 53.57 | 57.15 | 46.65 ± 10.47 |
| Test | Images | TAO-Net | 36.84 | 52.63 | 63.64 | 59.70 | 63.75 | 55.31 ± 10.08 |

**Table A2.** F1 score for each class in the test set for the multi-class detection. The last column reports the average F1 score of the six classes.

| Set | Data | Model | BZ | AGN | CV | OTHER | SN | Non-transient | F1($\mu \pm \sigma$) |
|---|---|---|---|---|---|---|---|---|---|
| Test | Images | TAO-Net | 38.46 | 34.21 | 53.19 | 41.12 | 46.63 | 95.46 | 51.51 ± 20.56 |

**Table A3.** F1 score for each class in the test set for the binary problems. The last column reports the average F1 score of both classes.

| Set | Data | Model | Transient | Non-transient | F1($\mu \pm \sigma$) |
|---|---|---|---|---|---|
| Test | Images | TAO-Net | 88.38 | 96.62 | 92.50 ± 4.12 |
| Set | Data | Model | Supernova | Non-transient | F1($\mu \pm \sigma$) |
| Test | Images | TAO-Net | 75.52 | 97.50 | 86.51 ± 10.99 |

This paper has been typeset from a TEX/LATEX file prepared by the author.



Downloaded from https://academic.oup.com/mnras/article/499/3/3130/5917433 by guest on 15 October 2024